Rocket Propulsion Simulator
Technical Evidence & Resume Assessment

October 16, 2026 | Project: Two-Stage Sounding Rocket Flight and Nozzle Performance Simulation
Repository: github.com/SJF-Aero/Rocket-Flight-Simulator

Executive assessment

The strongest defensible accomplishment is the development of separate Python models for two-stage vertical rocket flight and ideal compressible nozzle flow, supported by numerical checks and sensitivity studies. The revised flight case predicts 258.0 km apogee; an independent RK4 calculation differs by about 0.04%. This measures numerical agreement, not physical flight-prediction accuracy.

Evidence standard: The results below are recovered from earlier project documentation and conversations. They were not independently rerun for this report. Values labeled as simulated predictions must not be presented as measured flight or thrust results.

1. Implemented capabilities and scope

Component	Completed capability	Evidence status
Flight.py	1-D vertical ascent and coast; thrust curves; propellant depletion; stage separation; altitude-dependent gravity and atmosphere; Mach-dependent drag; dynamic pressure; variable mass.	Documented as implemented
Nozzle.py	Separate ideal isentropic nozzle solver: choked mass flow, area-Mach inversion, exit conditions, thrust, specific impulse, characteristic velocity, thrust coefficient.	Documented as implemented
Test_Flight.py	Reported 13 passing tests including impulse, atmosphere, drag sign, burnout mass, propellant conservation, liftoff, analytical comparison, timestep behavior, and nozzle behavior.	Reported, not rerun here
Not implemented	Nozzle-to-flight coupling; 6-DOF attitude; wind; Earth rotation; physical nozzle tests; manufactured nozzle geometries.	Future or out of scope

2. Flight results: revised reference configuration (Case B)

Metric	Documented result	Interpretation
Predicted apogee	258,014 m (258.0 km)	Numerical model output, not validated flight altitude
Max dynamic pressure	575 kPa at ~10.4 s, ~8.64 km, Mach 5.03	Model output using assumed aerodynamics
First-stage burnout	4.4 s; ~1.71 km; ~822 m/s	Model event
Second-stage burnout	28.4 s; ~39.1 km; ~2,029 m/s	Burnout speed, not proven maximum speed
Apogee time	~247 s	Model event
First-stage input	257.5 kN for 4.4 s	Provisional motor approximation
Launch mass	~1,565 kg	Model configuration
Maximum velocity / acceleration	Not established as exact extrema	Must extract maxima from full time histories

Independent numerical integration

Configuration	Primary solver	Independent RK4	Difference
Revised B	258,014 m	258,126 m	112 m; 0.0434%
Original A	126,987 m	127,015 m	28 m; 0.0220%

Relative difference = |h(RK4) - h(primary)| / h(RK4) x 100%. Revised case: 112 / 258,126 x 100% = 0.0434%. This establishes agreement of numerical integrations using shared model assumptions, not 0.0434% accuracy against flight measurements.

Timestep refinement

Timestep	Predicted apogee
0.040 s	257,677 m
0.020 s	257,901 m
0.010 s	258,014 m
0.001 s	258,115 m

Refining 0.010 s to 0.001 s changes apogee by 101 m, or 101 / 258,115 x 100% = 0.0391%. This is an observed convergence difference, not a formal numerical-error upper bound.

3. Four-configuration sensitivity analysis

Case	First stage	Second stage	Launch mass	Apogee	Max-Q
A	Old Terrier	Model Orion	1,429 kg	127.0 km	427 kPa
B	New Terrier	Model Orion	1,565 kg	258.0 km	575 kPa
C	New Terrier	NASA-sheet Orion	1,451 kg	191.7 km	616 kPa
D	Old Terrier	NASA-sheet Orion	1,315 kg	80.7 km	327 kPa

Apogee spans 177.3 km (258.0 - 80.7 km) across provisional input configurations. A +/-20% change in assumed drag coefficient yielded approximately 237-281 km for Case B. In the original configuration, a 4-6 s interstage coast reduced Max-Q from ~427 kPa to ~365-383 kPa with less than 1 km apogee change. These are modeled sensitivity outcomes, not verified vehicle improvements.

The NASA-sheet Orion case uses published approximate thrust levels but inferred propellant mass and an assumed dry mass selected to approximate a published vehicle total. Agreement with a broad published altitude range cannot be treated as independent validation.

4. Ideal nozzle model: documented reference operating point

Input	Value	Output	Value
Chamber pressure	3.0 MPa	Choked mass flow	3.498 kg/s
Chamber temperature	3,400 K	Exit Mach number	3.173
Heat capacity ratio	1.22	Exit pressure	48.10 kPa
Gas constant	355 J/(kg K)	Exit velocity	2,652 m/s
Throat diameter	50 mm	Sea-level thrust	8,441 N
Expansion ratio	8	Sea-level specific impulse	246.1 s
Ambient pressure	Sea level	Characteristic velocity	1,684 m/s
		Thrust coefficient	1.433
		Optimum sea-level expansion ratio	~4.66
		Optimum sea-level thrust	~8,653 N

Core equations: F = m_dot x v_exit + (p_exit - p_ambient) x A_exit; Isp = F / (m_dot x g0). The calculated nozzle output is idealized; no flow separation, viscosity, combustion losses, or divergence losses are modeled. The nozzle is not coupled to the prescribed flight thrust curves.

5. Validation versus verification

Evidence	Supported claim	Not supported
Independent RK4: ~0.04% apogee difference	Numerical solver agreement	0.04% real-flight accuracy
Timestep refinement: ~0.04% change	Observed convergence	Formal error bound
Atmospheric reference comparison: reported within 0.02%	Agreement with selected atmosphere references	Validation of full rocket dynamics
13 reported passing tests; simplified analytical rocket-equation test with 1 m/s tolerance	Selected software and physics checks	Complete model validation
NASA broad payload/apogee performance description	Useful external plausibility comparison	Matched-condition flight validation
No physical tests	No experimental claim	Measured nozzle thrust or flight results

NASA comparison limitations

The report cites an approximate NASA Terrier-Improved Orion performance envelope of ~80 km at 800 lb payload to ~200 km at 200 lb payload. Case B predicts 258 km for a 300 lb payload. Comparing 258 to 200 yields an illustrative 29% excess, but 29% is not a defensible flight-prediction error: payloads, motor configurations, masses, aerodynamics, and launch conditions do not match. The NASA chart reportedly assumes 85-degree launch elevation; the simulator assumes vertical ascent. Some input data are provisional or unconfirmed.

A defensible validation percentage requires a matched reference flight or authoritative configuration-specific trajectory, verified motor and vehicle parameters, and the same comparison metric.

6. Top five resume-worthy accomplishments

Rank	Accomplishment	Assessment
1	Developed 1-D two-stage flight simulator with atmosphere, variable mass, drag, and staging	Strongest engineering complexity; implemented
2	Independent RK4 agreement of 0.0434% in apogee	Strong numerical-verification metric; not physical accuracy
3	Four-case sensitivity study, 80.7-258.0 km apogee	Demonstrates engineering judgment about uncertain inputs
4	Ideal compressible-flow nozzle performance model	Strong propulsion fundamentals; outputs theoretical
5	13 reported automated tests and timestep convergence study	Strong software-quality evidence; rerun tests before claiming current passing status

7. Recommended resume bullet points

- Developed Python-based two-stage rocket flight and nozzle simulators incorporating compressible flow, variable-mass dynamics, atmospheric modeling, aerodynamic drag, and stage separation.

- Numerically verified flight predictions using independent RK4 integration and timestep convergence analysis, achieving 0.04% agreement in predicted apogee between solvers.
- Conducted four-configuration propulsion sensitivity analysis, evaluating predicted apogees from 80.7-258.0 km and identifying motor and mass assumptions as dominant sources of model uncertainty.

If space permits only two bullets: Use the first two. Do not claim experimental validation, measured 8.44 kN thrust, NASA-validated accuracy, or an achieved 258 km flight.

8. Compact evidence summary

Metric / accomplishment	Actual value	Evidence	Status / resume use
Two-stage flight model	1-D, two stages	Flight.py; report sections 1-4	Implemented; yes
Ideal nozzle model	Isentropic, choked flow	Nozzle.py; report section 4	Implemented; yes
Revised apogee	258,014 m	Report sections 5-6	Predicted; qualify
RK4 comparison	0.0434% difference	Report section 5.2	Numerical agreement; yes
Timestep convergence	101 m / 0.0391%	Report section 5.2	Observed change; yes
Automated tests	13 passing reported	Test_Flight.py; report section 5.1	Rerun recommended
Four-case apogee range	80.7-258.0 km	Report section 6.1	Sensitivity; yes
Max-Q range	327-616 kPa	Report section 6.1	Predicted; optional
Drag sensitivity	~237-281 km	Report sections 6.2-6.3	Predicted; optional
Burnout speed	~2,029 m/s	Report section 6.1	Predicted; optional
Nozzle thrust / Isp	8,441 N / 246.1 s	Nozzle output table	Idealized; optional
Max velocity / acceleration	Unavailable as exact peaks	Time-history extraction needed	Do not claim
NASA flight error	No defensible percentage	Report sections 2, 6.2	Not validated; no
Experimental nozzle testing	Not performed	Project future-work section	Planned; no

9. Next verification priorities

- Rerun the automated tests from the published repository and preserve the exact command and output.
- Make the RK4 comparison and timestep convergence results reproducible from the published code.
- Extract maximum velocity and acceleration directly from recorded flight histories, checking staging discontinuities and pad-contact logic.
- Acquire independently verified motor thrust curves, masses, and aerodynamic inputs; compare a matched reference flight before claiming flight validation.
- Keep CAD, nozzle manufacturing, and cold-flow thrust testing labeled as future work until completed.

Source and confidence note

Basis: recovered versions of the project technical report titled Two-Stage Sounding Rocket Flight and Nozzle Performance Simulation, accompanying project notes, and previously documented simulation results. This is a resume-evidence synthesis, not an independent audit of the current GitHub source files or a new execution of the simulator. NASA source descriptions are reported as cited within the project report, not freshly authenticated in this document.